

FADIL KARIM

AI ENGINEER



SUMMARY

Expert in Revit, AutoCAD, and Bentley Suites. Specialist in Computational Design with Grasshopper, Dynamo, and Python. Experienced in building AI agents for Revit. Proficient Python coder with a deep understanding of data management & structures. Skilled in implementing AI into BIM workflows, using AI to make live changes to and analyse Revit models. Creator of a YouTube channel with 7k+ subscribers where I teach BIM modelling with Revit.

CHANNELS



OttomanLabs.AI YouTube Channel
Practical AI for Revit Tutorials
[YouTube.com/@OttomanLabsAI](https://www.youtube.com/@OttomanLabsAI)

website: OttomanLabs.AI

SOFTWARE SKILLS

3D BIM



COMPUTATIONAL DESIGN AND CODING



GRAPHIC DESIGN



EDUCATION

2021 - 2023

- King's College London
Applied Neuroscience, PGCert

2010 - 2014

- University of Birmingham
Civil Engineering, BEng

2008-2010

- St. Joseph's College
A-Levels: Maths (A), Further Maths (B), Physics (C)

EXPERIENCE SUMMARY

SECTOR EXPERIENCE

Building Structures | 4 yr 11 mo
Water Infrastructure | 4 yr 1 mo
MEP | 1 yr 8 mo
Rail | 1 yr 2 mo
Temporary Structures | 8 mo

ROLE EXPERIENCE

BIM Manager | 1 yr 3 mo
AI Engineer | 6 mo
Senior BIM Technician | 8 yr 2 mo
BIM Technician | 3 yr 1 mo

TECHNICAL EXPERIENCE

BIM for Construction

3D Modelling - Revit, MicroStation/AECOSim, Civil 3D, AutoCAD

- In-depth knowledge of BIM data.
- 13+ years experience building 3D models of Structures (Buildings, Rail), MEP (Lighting, Drainage pipes, Electrical ducts), and Architecture (Rail).
- 7+ years experience building Parametric Revit Families.
- 13+ years experience creating drawings for Structures, MEP & Architecture.
- 5+ years experience in developing automation for BIM Modelling.

BIM Management

- Managed and coordinated BIM Technicians and BIM Coordinators.
- Developed BIM Execution Plans for MEP Lighting and Power Cables.
- Developed and upheld project standards, workflows, and information management procedures in accordance with BS EN ISO 19650 and UK BIM standards.
- Led BIM coordination activities including clash detection, model federation, design reviews, and issue resolution across disciplines.
- Oversaw the management and exchange of project data between multiple software platforms and Common Data Environments (CDE).
- Liaised with clients, consultants, and stakeholders to ensure alignment of BIM deliverables with project objectives.

DATA ENGINEERING for BIM

Automation - Python, C#, Dynamo & Grasshopper

- Deep understanding of Data & Information Management.
- Expert Python, Dynamo and Grasshopper.
- Able to integrate all CAD/data file types into a single model.

I developed the following automations to cut days of work down to minutes. All were shared with other team members used them without the need for expertise or additional software licenses.

Structures

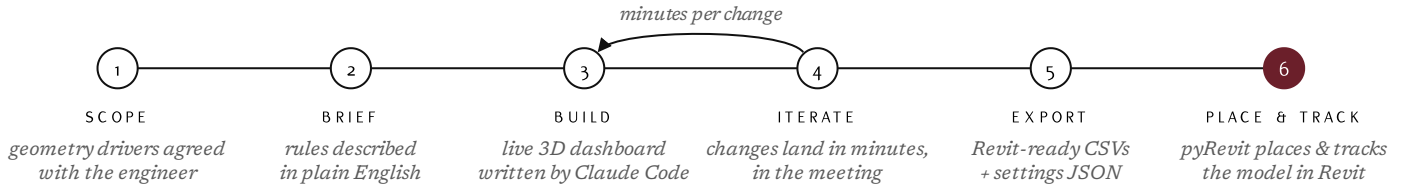
- Extend/Align bottom structural elements to topography.
- Create & place hundreds of framing for complex structures.
- Place & size columns/beams directly from engineer's raw excel files.
- Create hundreds of named and numbered sheets in seconds.
- AI Agents for Revit API for 3D modelling via text prompt.
- AI Agents for spacial analysis of CAD (.dwg) files via text prompt.
- Automated optimised warehouse design in Grasshopper & Dynamo.

MEP

- Place and size fence & column elements on 2D model lines.
- Create concrete encasements for thousands of cable ducts runs.
- Place 3D manholes and pipes in Revit from client's .dwg files.
- Populate tens of thousands of COBie parameters in multiple models.
- Place thousands of lights directly from CAD (.dwg) files.
- Annotate hundreds of pipe slopes & diameters with custom text formats in a single click with pyRevit custom buttons.

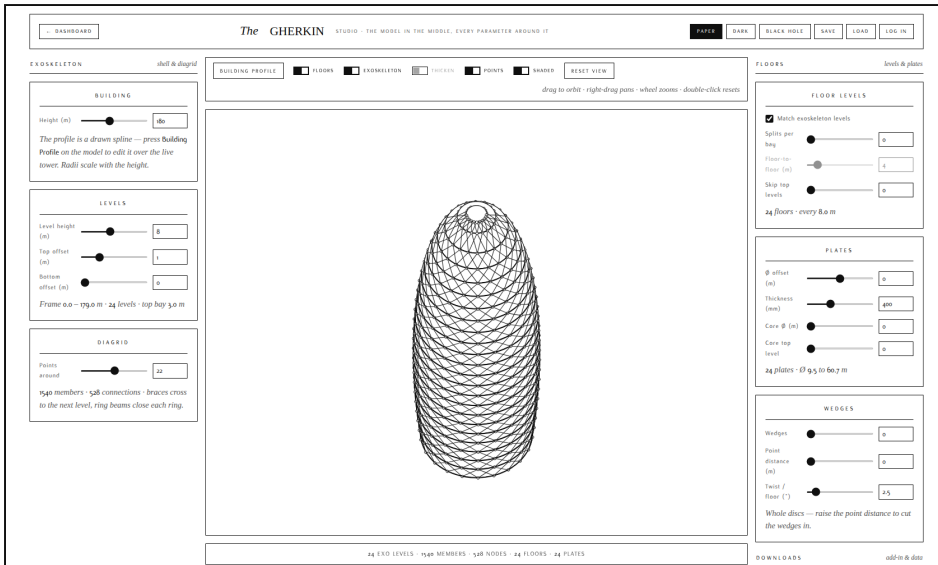
COMPANY HISTORY

Glent Group | Senior BIM Technician | Contract | Mar 2026 – Present
Amazon | AI Engineer | Contract | Oct 2025 – Mar 2026
EGIS Group | BIM Manager | Contract | Dec 2024 – Oct 2025
Pell Frischmann | Senior BIM Technician | Contract | Sep 2024 – Dec 2024
Whitfield Consulting | BIM Manager | Contract | May 2024 – Aug 2024
ES Global | Senior BIM Technician | Contract | Jan 2024 – May 2024
Eckersley O'Callaghan | Senior BIM Technician | Contract | Oct 2023 – Jan 2024
Buro Happold | Senior BIM Technician | Contract | Nov 2022 – Sep 2023
GHD | Senior BIM Technician | Contract | Jan 2022 – Nov 2022
Robert Bird Group | Senior BIM Technician | Contract | Aug 2020 – Dec 2020
3DReid | Senior BIM Technician | Contract | Jun 2020 – Aug 2020
Robert Bird Group | Senior BIM Technician | Contract | Oct 2019 – Apr 2020
BG&E | Senior BIM Technician | Full-time | Feb 2019 – Oct 2019
Arcadis | BIM Technician | Full-time | Jan 2018 – Feb 2019
AECOM | BIM Technician | Full-time | Feb 2016 – Dec 2017



THE GHERKIN STUDIO

built in one conversation – the model in the middle, every parameter around it



HOURS, NOT WEEKS

A working tool with a proper UI from the first conversation – the Grasshopper / Dynamo equivalent takes weeks, and years of experience to build well.

ZERO LICENCE COST

Runs in any browser; pyRevit is free. No Rhino, no add-on stacks, no training programme for the team.

ANYONE CAN DRIVE IT

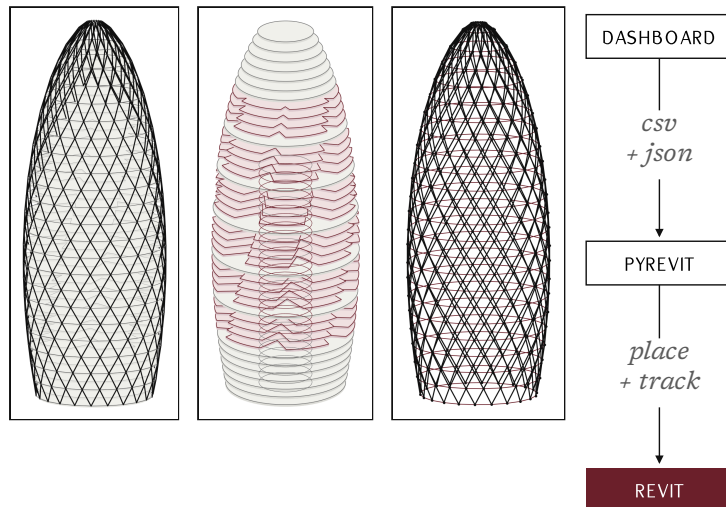
It is a web page, not a node graph – good enough UX to hand to the whole team on day one.

One browser tab: spline profile, exoskeleton levels, diagrid density, plates, wedges, core and twist – rebuilt live on every slider move.

HEIGHT 180 M · LEVELS 24 · MEMBERS 1,540 · CONNECTIONS 528 · PLATES 43

MODEL HAND-OFF

one click, Revit-native



One click exports the full model – and the same files let any Claude chat rebuild it exactly and render presentation views on demand. These images were generated that way.

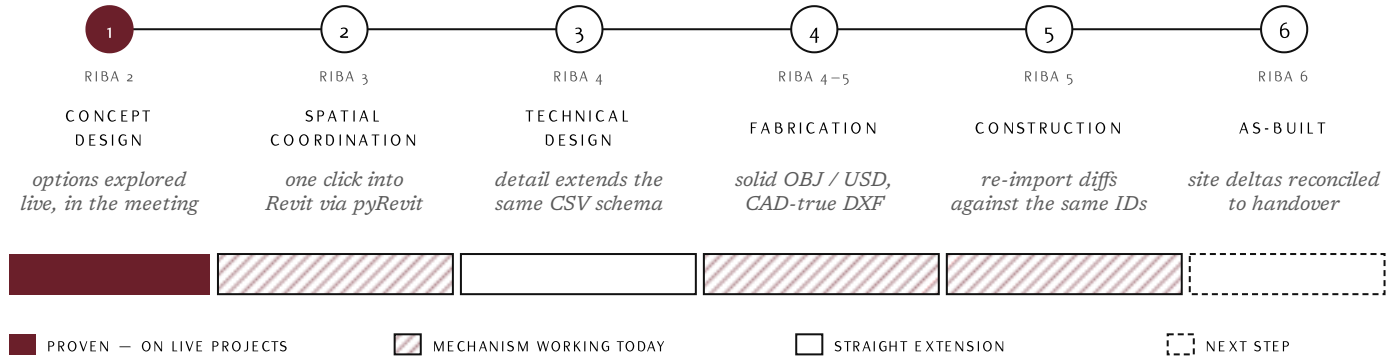
VS GRASSHOPPER & DYNAMO

same destination

	CLAUDE CODE	GRASSHOPPER / DYNAMO
BUILD TIME	Hours to a finished, shareable tool	Weeks of graph building
TRAINING	None – plain English	Years of specialist experience
LICENCES	Browser + free pyRevit	Rhino / add-on stacks
USERS	Anyone with the link	Trained specialists only
OUTPUT	The same Revit-native geometry – placed and tracked via pyRevit	

A fraction of the time and cost – and the tool itself is good enough to hand to the whole team on day one.

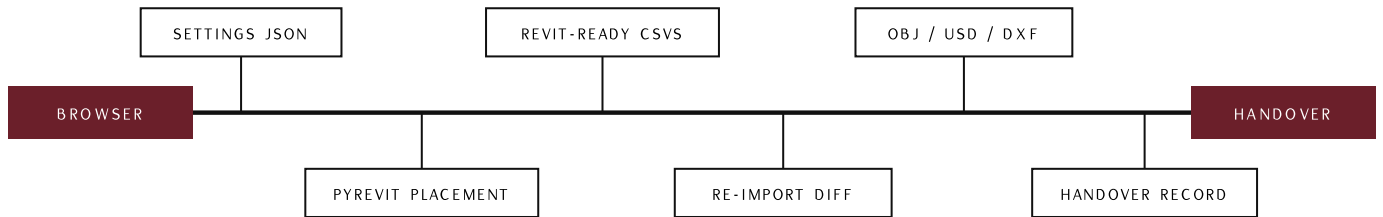
In use on **live projects** at concept design today – **project setup is already faster**. Built to carry the same model straight through **detailed design and beyond**. Live at **OttomanLabs.AI**.



the strength today is concept design – every mechanism the later stages need is already running underneath it

ONE MODEL, ONE SET OF IDS

every stage adds data to the same spine instead of starting again



stable tracking IDs, end to end – the concept model is already the delivery model

PROVEN TODAY

concept design, on live projects

THE ROUTE FORWARD

detailed design and beyond

Not a demo – this workflow runs on **live projects** today, and **project setup is already faster** because of it. A working parametric tool with a proper UI **within hours of the first conversation**, every change landing **in the meeting, not between meetings** – at zero licence cost, driveable by anyone on the project. Nothing is throwaway: **the concept model is already the delivery model**, exports included.

REVIT-NATIVE

Concept output already lands in Revit – pyRevit places levels, framing and plates today, with stable IDs from the first slider move.

SCHEMA, NOT REBUILD

Detailed design hangs connections, sections and specs off IDs that already exist – an extension, not a new model.

CHANGE CONTROL

Re-import diffs the model – moved, new, deleted – so coordination and site changes are tracked, not re-drawn.

ISO 19650-SHAPED

The settings JSON is a one-file audit trail – named deliverables, versioned states, CDE-native from day one.

DAYS → HOURS TO A WORKING STUDIO · DAYS → MINUTES PER DESIGN CHANGE · £0 LICENCE COST · 1 CLICK TO REVIT

The sweet spot today is concept design – but the pipeline is Revit-native and ID-tracked from day one, so every later stage extends the same model rather than starting a new tool. Extending into a new stage needs prompts, not products.